Nanyang Technological University
School of Electrical & Electronic Engineering
IE2106 Engineering Mathematics I

Tutorial 2 Probability & Statistics

1. A chip is drawn from a box containing 4 chips labelled 000, 011, 101, 110. Let A, B, C be the following events:

   A = the 1st digit on the drawn chip is 1,

   B = the 2nd digit on the drawn chip is 1,

   C = the 3rd digit on the drawn chip is 1.

   Write down the 3 sets that represent the 3 events.

   Show that

   (a) events A and B are independent,

   (b) events A and C are independent,

   (c) events B and C are independent,

   (d) events A, B and C are dependent.

2. Consider the population of all families with 3 children. Let G denote a girl and B denote a boy. Assume that a child is equally likely to be male or female. (As an example, the notation BGB means the first child is a boy, the second is a girl and the third is a boy.)

   (a) Write down all the possible children compositions for a family with 3 children.

   A family is selected at random from the population.

   (b) What is the probability of obtaining a family with exactly 2 girls, given that the family has at least one girl?

   (c) What is the probability of obtaining a family with only 1 girl, given that the eldest child of the family is a boy?

3. Let X be the lifetime of a battery in hours. The random variable X has the following probability density function

$$f(x) = \begin{cases} \frac{1}{200}\, e^{-x/200}, & x > 0 \\ 0, & otherwise. \end{cases}$$

Find the probability that the lifetime of a particular battery

(a) is less than 200 or greater than 400 hours,

(b) lasts more than 300 hours, given that it has already been in use for more than 200 hours.

Answers

2.(a) GGG, BGG, GBG, GGB, BBG, BGB, GBB, BBB (b) $\frac{3}{7}$ (c) $\frac{1}{2}$

3.(a) 0.767 (b) 0.607